Where Does the Investment Demand Shock Come From?

A toy model of idiosyncratic shocks and lumpy investment

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A toy model of idiosyncratic shocks and lumpy investment[1]

Why is it legitimate to use a reduced-form investment demand shock?

Standard New-Keynesian CES aggregator and demand:

$$Y_t = \left(\int_0^1 y_{jt}^{\frac{\eta-1}{\eta}} \, dj \right)^{\frac{\eta}{\eta-1}}, \qquad y_{jt} = \left(\frac{p_{jt}}{P_t} \right)^{-\eta} Y_t, \qquad y_{jt} = a_{jt} k_{jt}.$$

Linear-in-capital technology $y_{jt} = a_{jt} k_{jt}$. Combining demand and production, real revenue is

$$r_{jt}(a_{jt}, k_{jt}) = \frac{p_{jt} y_{jt}}{P_t} = (a_{jt} k_{jt})^{1-\frac{1}{\eta}} \, Y_t^{\frac{1}{\eta}}.$$

- Concave in own capital ($\eta > 1$): diminishing returns through the demand curve.
- Revenue, hence the desired capital, scales with aggregate output Y_t — the channel through which one firm's action will later spill over to all firms.

[1] Building on *Nirei and Ragot (2025)*

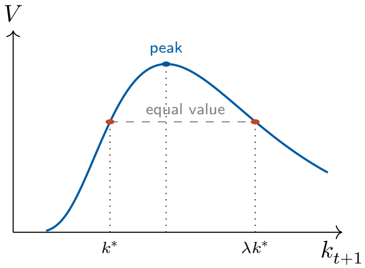
Lumpy investment and the investment threshold

Firms solve

$$\max_{\{x_{t+k}\}} \ \mathbb{E}_t \sum_{k \geq 0} \Lambda_{t,t+k}(r_{j,t+k} - x_{j,t+k}),$$

$$\text{s.t.} \ k_{j,t+1} = (1-\delta)k_{j,t} + x_{j,t},$$

$$x_{j,t+k} \in \{0, \ \lambda k_{j,t+k}\}.$$



Indifference at k^* (the firm knows its next-period technology) pins down the threshold:

$$k^*_{j,t+1} = a^{\eta-1}_{j,t+1}\, \Phi_t(\Lambda_t, \Lambda_{t+1})\, Y_{t+1}$$

- Desired capital scales with aggregate output Y_{t+1} — the spillover channel that will carry one firm's action to all firms.

Normalize a firm's position by its **distance to the threshold**

$$s_{jt} = \frac{\log(k_{jt}/k^*_{jt})}{\log \lambda},$$

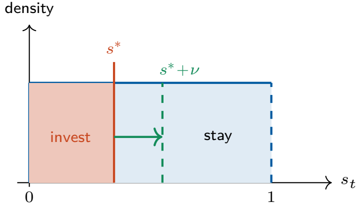
so that investing means jumping up exactly one notch ($s \mapsto s+1$) in these units.

Investing fraction and idiosyncratic shock

- Assume $a_t \in \{a_L, a_H\}$ and $s_t \mid a_t \sim U[0,1)$ under stationary distribution.
- Macro variables are constant: $Y = Y_t = Y_{t+1}$, $\Phi = \Phi_t = \Phi_{t+1}$

Without investing, under stationary distribution position drifts by

$$s_{j,t+1} = s_{j,t}(a) - \underbrace{\frac{-\log(1-\delta) + (\eta - 1)\log\frac{a'}{a}}{\log\lambda}}_{s^*(a,a')}.$$



- Since $s_t \sim U[0,1)$, $s^*(a,a')$ is the **investing fraction**
- Add an i.i.d. shock ε on a' *only* in group $(a,a') = (b,b')$:

$$s^*(b, \varepsilon b') = s^*(b,b') + \underbrace{\frac{(\eta-1)\log\varepsilon}{\log\lambda}}_{\nu}.$$

- The cutoff in that group rises by ν, so an extra ν-mass invests.

Propagation through the threshold

More investment in (b, b') raises *future output* of this group, which is a part of *future aggregate output*:

	$a' = L$	$a' = H$
$a = H$	$\omega(H, L)$	$\omega(b, b')$
$a = L$	$\omega(L, L)$	$\omega(L, H)$

$$Y_{t+1}(a, a', \nu) = \left(\omega(b, b') \int_0^{s^*+\nu} \left(b'\lambda(1-\delta)k_t(s_t)\right)^{\frac{\eta-1}{\eta}} dF(s_t \mid b) + \text{no investment firms}\right)^{\frac{\eta}{\eta-1}}$$

- Higher Y_{t+1} shifts the cutoff in *every* group:

$$\nu^{(1)}(a, a') := \Delta s_t^*(a, a', \nu)$$
$$= s^*(a, a') + \frac{\log(Y_{t+1}(\nu)/Y)}{\log \lambda} - \left(s^*(a, a') + \frac{\log(\overset{=Y}{Y_{t+1}(0)}/Y)}{\log \lambda}\right)$$

- aggregate ν-**induced fraction**:

$$\nu^{(1)} := \sum_{(a,a')} \omega(a, a')\, \nu^{(1)}(a, a').$$

- The first ν of investing firms raises Y, which induces $\nu^{(1)}$ more, which raises Y again ... a GE multiplier that iterates to a new equilibrium.

Total induced investment $\gg \nu$

$$\text{Total induced investment} = \sum_{i=1}^{\infty} \text{induced fraction}^{(i)} \times \text{average induced investment}^{(i)}$$

- Scale of induced fraction

$$\lim_{\nu \to 0} \frac{\nu^{(1)}}{\omega(b, b')\, \nu} = 1$$

  - the $\nu^{(1)}$ has the *same scale* as the original fraction
    $\Rightarrow$ each $\omega(a, a')\nu^{(1)}(a, a')$ will induce the *same scale* $\nu^{(2)}(\nu^{(1)}(a, a'))$ again
    $\Rightarrow$ induced fraction will not decay through iteration.

- Scale of average induced investment

$$\left.\frac{\partial \Delta x(a, a', \nu)}{\partial \nu}\right|_{\nu=0} = (\lambda - 1)(1 - \delta)\lambda^{s^*} a^{\eta-1} \Psi Y$$

  - ν-induced average investment, $\Delta x(a, a', \nu)$, is defined as
    $\Delta x_t(a, a', \nu) := \int_{s^*}^{s^*+\nu} (\lambda - 1)(1 - \delta) \underbrace{\lambda^{s_t} a^{\eta-1} \Phi Y}_{k_t(s_t)} \, dF(s_t \mid a_t = a)$
    $\Rightarrow$ the derivative says $\Delta x(a, a', \nu) = O(\nu)$
    $\Rightarrow$ average induced investment will not decay faster than ν.

Takeaways

1. **Propagation** $=$ a within-group idiosyncratic shock raising aggregate output, which feeds back onto every firm's investment threshold.
2. **Lumpy investment** is essential: it creates the threshold (the investing fraction s^*) and guarantees each induced investment is large, so the aggregate effect does not wash out.
3. **Reduced-form ID shock** is justified: the aggregated micro shock survives as a degenerate shock on investment, validating adding it directly to investment in estimation.